MODEL

(Revision – 2015) Reg. No. ……………………

TED(15): 5111 Signature. …………………..

FIFTH SEMESTER DIPLOMA EXAMINATION IN TOOL AND DIE ENGINEERING

PRESS TOOL TECHNOLOGY

[Maximum Marks: 100] [Time : 3 Hour]

PART – A

[Maximum Marks : 10]

I. Answer the following question in one or two sentences: Marks

1. Define lancing operation
2. What is an indirect pilot?
3. Name different types of set screws used in press tools
4. Write two advantages of coining operation
5. What is a compound die? [5x2=10]

PART – B

[Maximum Marks: 30]

II. Answer *any five* of following questions. Each question carries 6 marks

1. Explain about cutting and non cutting punches
2. Sketch and explain buckling of punches
3. What are the advantages of side cutting tool?
4. Calculate the bending force for a 45^0 bend in aluminium blank. The following data is given:

   Blank thickness = 3 mm; Bend length = 2000 mm; Die opening = 8 x metal thickness

   Ultimate tensile strength = 500N/mm^2.
5. List out advantages and limitations of compound tool
6. Describe about planishing operation
7. A shell having 40 mm diameter and 120 mm height with a wall thickness of 1 mm is to be manufactured from 1.3 mm metallic sheet. Calculate the blank diameter

[5x6=30]

PART – C

[Maximum Marks: 60]

[Answer one full question from each unit. Each full question carries 15 marks]

UNIT - I

III. (a) A steel component of 25mm x 62mm is to be made from 2mm thick sheet. Sketch the scrap strip layout and also determine the percentage of stock used. 8

(b) What is a clearance? Why it is important in shearing operation 7

OR

IV. (a) State the function of stoppers, describe anyone in detail 8

(b) Explain the shearing theory of metals between die components 7

UNIT - II

V. (a) A steel washer of 60mm outer diameter and 30mm inside diameter is to be cut from 1.6mm thick sheet in one operation. If the shear stress is 500 N/mm^2 determine the maximum punch force necessary to blank and punch the washer if both punches operates at one time 8

(b) What is a "stock stop" and "pilot" 7

OR

VI. (a) Two holes, one 5 cm x 4 cm rectangle and another 10cm diameter are to be cut in a metallic sheet of 3mm thick. If shear strength of material is 4000 kg/cm^2, determine

(i) Cutting force (ii) Stripping force 8

(b) Explain with a neat sketch about fixed stripper 7

UNIT – III

VII. (a) What is bending of materials? Sketch and explain various bend elements 8

(b) Explain about materials used for bending 7

OR

VIII. (a) Explain working of progressive die with a neat sketch. 8

(b) State the effect of grain direction in bending 7

UNIT – IV

IX. (a) sketch and explain drawing tool using pressure pad 8

(b) Discuss material flow in fine blanking 7

OR

X. (a) Explain about following forming operation

(i) Embossing (ii) Flanging (iii) Drawing 8

(b) Explain about fine blanking operation 7